

Project I: Analysis of a Classifier

Afonso Vilaça
Image Analysis
Chalmers University of Technology
Gothenburg, Sweden
vilaa@chalmers.se

Abstract—This project addresses the challenge of automated classification of bone marrow cells. A baseline Convolutional Neural Network (CNN) was developed to classify $56 \times 56 \times 3$ RGB images into eight distinct cell categories. While the baseline architecture achieved a test accuracy of approximately 70%, confusion matrix analysis revealed significant misclassification between similar classes, specifically the NGS and the NGB stages. To improve how the model generalizes and feature extraction, three targeted methodologies were integrated: Data Augmentation (random geometric transformations) to prevent overfitting, Batch Normalization layers to accelerate convergence, and Early Stopping logic monitored via a subset of the training data to prevent overfitting. The optimized model demonstrated substantial performance gains, achieving a final test accuracy of approximately 82%.

I. MANDATORY PART

A. Method

1) **Baseline CNN:** The initial baseline model was designed as a simple Convolutional Neural Network (CNN) to classify $56 \times 56 \times 3$ RGB images into eight distinct bone marrow cell classes. The architecture consists of two convolutional layers (with 16 and 32 filters, respectively, using 3×3 kernels), each followed by a ReLU activation function and a 2×2 max-pooling layer to reduce spatial dimensions. The flattened output is then passed through two fully connected layers (with 128 and 8 nodes) to generate the final class predictions. The detailed structure can be found below:

- 1) **Input:** $56 \times 56 \times 3$ (RGB Image)
- 2) **Layer 1 (Conv + ReLU + Pool):**
 - a) **Conv1:** Kernel 3×3 , Stride 1, Padding 1, Filters 16. (Output size: 56×56)
 - b) **MaxPool1:** Kernel 2×2 , Stride 2. (Output size: 28×28)
- 3) **Layer 2 (Conv + ReLU + Pool):**
 - a) **Conv2:** Kernel 3×3 , Stride 1, Padding 1, Filters 32. (Output size: 28×28)
 - b) **MaxPool2:** Kernel 2×2 , Stride 2. (Output size: 14×14)
- 4) **Layer 3 (Flatten+ Linear+ Relu + Linear):**
 - a) **Flatten:** Takes the 3D grid of features from the previous layer and unrolls it into a single, 1D array so the standard neural network layers can read it (Output Dimension: 6,272 nodes ($14 \times 14 \times 32$))
 - b) **Linear1:** Connects the 6,272 input nodes to 128 hidden nodes

- c) **Linear2:** Connects the 128 hidden nodes down to the final 8 output nodes. ((Output Dimension: 8 nodes) (Each representing the raw prediction score for one of the 8 bone marrow cell classes))

2) **Selected Improvements:** Based on an initial evaluation of the baseline model, which achieved approximately 70% test accuracy but struggled significantly to differentiate between certain classes (specially NGS vs. NGB and BLA vs. PMO), three specific improvements were implemented to enhance feature extraction and training stability

- 1) **Data Augmentation:** The baseline confusion matrix indicated the model was likely overfitting to the spatial orientation of the cells (particularly confusing NGS and NGB). To make the network learn orientation-invariant biological features, random horizontal flips, random vertical flips, and random rotations (up to 45 degrees) were applied to the training dataset
- 2) **Batch Normalization:** To stabilize the learning process and mitigate internal shift, *BatchNorm2d* layers were inserted after each convolutional layer, and a *BatchNorm1d* layer was added before the final fully connected layer. This smoothed the loss and allowed the network to learn the differences between similar classes like BLA and PMO much faster.
- 3) **Early Stopping:** Analysis of the training loss curve indicated the model required more than the initial 15 epochs to converge, introducing the risk of overfitting. A 20% validation split was made out from the training data. An early stopping monitor was implemented to track the validation loss at the end of each epoch. If the validation loss failed to improve for 5 consecutive epochs, training was stopped, and the weights from the optimal epoch were restored.

B. Experimental Evaluation

1) **Training Parameters:** The models were trained using the Adam optimizer to allow for adaptive learning rates. The initial learning rate was set to 0.001. Additionally, I added an L2 weight decay of $1e-4$ to provide regularization. To fine-tune convergence, a *ReduceLROnPlateau* scheduler was implemented, configured to divide by 2 the learning rate if the validation loss plateaued for 2 consecutive epochs. All training was conducted using Cross-Entropy Loss with a batch size of 64.

2) *Visualization of Improvements*: The implementation of Early Stopping and the Learning Rate Scheduler was important to achieve convergence without overfitting.

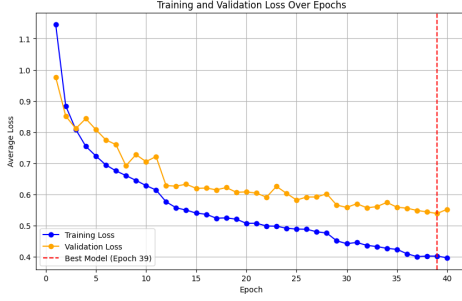


Fig. 1. Loss vs Epochs for Validation and Training sets

As shown in the figure above, the training loss decreased over 40 epochs. However, the validation loss stabilized and reached its absolute minimum at Epoch 37. The Early Stopping monitor correctly identified this inflection point, stopping training and restoring the Epoch 37 weights to prevent the network from memorizing the training data. Additionally, Data Augmentation was used during the training phase, utilizing 50% random horizontal and vertical flips, alongside $\pm 45^\circ$ random rotations, to successfully distinguish the cell classifications from their spatial orientation.

3) *Ablation Study Results*: The ablation study confirms the individual necessity of each engineered improvement. Reverting to base transformations (Model A) resulted in the most significant performance degradation (72.22%), validating that Data Augmentation is the primary driver of model generalizability. The removal of Batch Normalization (Model B) gave a test accuracy of 79.39%, demonstrating that normalization is required to achieve high-precision classification in complex tasks. Finally, removing the Early Stopping constraint (Model C) resulted in a test accuracy of 82.66%. While this slightly exceeded the fully improved model's 82.43% on the static test set, it highlights that while the model continued to learn in the final epochs, the validation loss was approaching a point of diminishing returns. The Improved Model (82.43%) remains the most robust configuration, as it explicitly protects against the degradation of generalization capability that would inevitably occur with extended, unmonitored training. The confusion matrices for each of the models can be seen in Figure 2:

4) *Subset Visualization*: I visualized some of the images, and compared their predicted and true label, as seen below:

- 1) **Top-Left Image** (True: ART, Predicted: NGB): The model misclassified this artifact as a NGB. This likely occurred because the purple nuclear material is stretched and slightly pinched in the middle, falsely resembling the typical 'U' or 'C' band shape characteristic of the NGB class.
- 2) **Top-Right Image** (True: ART, Predicted: BLA): This artifact was misclassified as BLA. Because this artifact is essentially a large, solid blob of dark purple nuclear

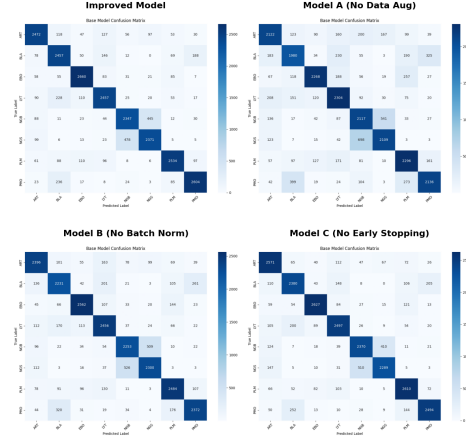
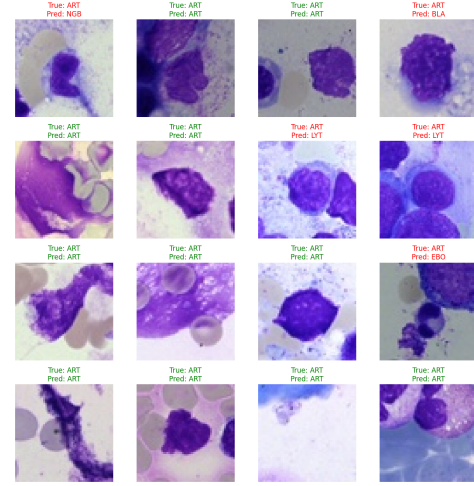


Fig. 2. Confusion matrices comparing the fully optimized model (Top Left) against the three ablation models. Note the significant increase in misclassifications between structurally similar classes, particularly NGS and NGB, when Data Augmentation is removed (Top Right).



material with no visible cytoplasm, the network's confusion is highly logical.

- 3) **Second Row, Third Image** (True: ART, Predicted: LYT): The model incorrectly predicted LYT. The artifact in this image settled into a tight, dark, round circle, perfectly mimicking the primary geometric features the CNN uses to identify LYT cells.

II. ADVANCED PART

A. Out-of-Distribution (OOD) Detection

To identify these images that the network was never trained on without retraining the model, the Maximum Softmax Probability (MSP) method was used. For each image passed through the network, the final logits were converted to probabilities via a softmax function, and the highest resulting probability was extracted to serve as a scalar confidence score.

The model was evaluated against the provided dataset of 2,000 mixed, unlabeled images. Because the images were not sorted into class folders, a custom PyTorch Dataset was engineered to process the flat directory and return the extracted

MSP scores alongside their corresponding file paths. The distribution of these scalar scores is visualized in the figure below.

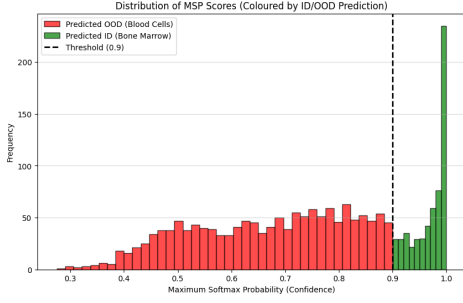


Fig. 3. Distribution of MSP scores

The resulting histogram demonstrates a clear bimodal distribution, successfully separating into two distinct sectors:

- 1) **The In-Distribution (ID) Sector:** The sharp concentration of scores approaching 1.0 (more than .90). By isolating the file paths in this range, it was confirmed that this peak represents the known bone marrow cells. The model outputs high confidence because these images map clearly to the parameters learned during training.
- 2) **The Out-of-Distribution (OOD) Sector:** The broader distribution of lower scores (around 0.3 to 0.9). Visual inspection of the file paths in this range confirmed they correspond to foreign blood cell images. Because these features lack structural similarity to the training set, the network's probability distribution remains flat across the 8 known classes, resulting in a distinctly lower maximum confidence score.

III. AI USAGE

I used AI to help me structure the CNN, so the output dimensions of each layer were correct. I also used it to help me with some LaTeX commands.